Facilities, Equipment and Other Resources

Georgia Institute of Technology

The project will use Georgia Tech research and web infrastructure for development, integration, and operation. CryoStack is currently deployed on a Georgia Tech Linux virtual machine behind Nginx and supports HTTPS, WebSocket-based applications, local-password/GitHub/ORCID authentication, role-based administration, persistent experiment records, and connection to remote computing resources. Institutional identity, networking, security, backup, and server-administration services will be used as approved by Georgia Tech. Ordinary faculty/student workstations support development, documentation, testing, and laptop-scale tutorial workflows.

Georgia Tech's Partnership for an Advanced Computing Environment (PACE) provides Slurm-managed high-performance computing, parallel filesystems, software modules, and user support. PACE has already been used for ICESEE MPI development and scaling experiments and will serve as the reference backend for adapter qualification and science demonstrations. Specific allocations, queues, storage quotas, and staff commitments will be confirmed before submission and described without assigning a monetary value as voluntary committed cost sharing.

Software and Deployment Resources

Existing project resources include the ICESEE ensemble data-assimilation framework; the CryoStack gateway, Control Center, and modular CryoLauncher web application; the LIVIST radar/borehole englacial-temperature explorer; the manifest-driven Frozen Legacies historical-radar catalog and its first LYRA adapter; desktop HPC connectors for Linux, macOS, and Windows; SSH/Slurm execution code; modular AWS identity, IAM, networking, storage, registry, and Batch components; Docker/OCI and Apptainer recipes; Spack packages and environments; model integrations for ISSM, Icepack, flowline, and Lorenz-96; and public documentation, tests, and issue tracking. The project also builds on PISM, PETSc, MPI, Firedrake, Jupyter Book, GitHub, and Zenodo, subject to their licenses and terms.

AWS services will be used only for bounded backend engineering, qualification, and demonstrations covered by the project budget, and for the bounded computing-access pilot if it is approved. CryoStack does not require AWS; outside these activities, cloud runs are billed by AWS to the researcher's own account. PACE remains the reference environment, while the AWS work establishes a documented path for project-managed and bring-your-own-account execution.

Collaborative Resources

The project will seek letters of collaboration documenting bounded technical review or user-evaluation roles from appropriate PISM users/developers, historical-radar data stewards, Georgia Tech PACE/web-security personnel, and research-software or science-gateway specialists. These collaborations will advise interface, security, usability, and portability decisions; they will not be represented as committed resources until the corresponding letters and institutional approvals are obtained.